

Article

A Blockchain-Enabled Smart Contract Architecture for Enhancing Transparency, Traceability, and Trust in Global Supply Chain Management

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Abstract

There is diminished transparency, fragmented information exchange, and lack of trust among geographically dispersed stakeholders, which increasingly challenge global supply chains. The classic centralized systems of supply chain management are not always capable of being able to offer real-time traceability and data integrity which is dependable and effective in contract enforcement. The proposed study is a blockchain-based smart contract design that is focused on ensuring increased transparency, traceability and trust in global supply chain management. The suggested framework will combine automated smart contracts, cryptographic provenance tracking, permissioned blockchain consensus, and a decentralized trust score evaluation mechanism to overcome some of the major operation and governance challenges. A simulated assessment with a multi-tier global supply chain setting of 15 blockchain nodes and 12,000 transactions was performed through experimentation. The findings show that the proposed system attained an average transaction delay of 210 ms, which is very low compared to centralized systems (520 ms), with throughput being raised to 120 transactions per minute. End-to-end traceability performance also improved significantly, with a reduction in trace-back time to 8 s compared with 95s this represents a 100% tampering detection rate. The consensus mechanism ensured that the ledger integrity failed only at a rate of less than 1.1%, even when more than 30% of nodes were faulty. Risk-wise, the trust evaluation algorithm dynamically enhanced reliable supplier scores up to 12%, which facilitated the selection of reliable partners. On the whole, the results prove that smart contracts based on blockchains can drastically enhance the efficiency of operations, data integrity, and confidence in global supply chains, with the platform capable of providing a resilient and scalable backbone for the future supply chain management model.

Keywords: blockchain; smart contracts; supply chain management; traceability; transparency



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1. Introduction

Supply chains have become very complex and networked and entail a wide range of stakeholders, geographic regions and regulatory environments across the globe. Although this globalization has made it possible to achieve cost efficiencies, an accelerated production process, and increased access to markets, it has also brought about some enduring issues in the form of a lack of transparency, limited traceability, silos, fraud, and lack of trust between the participants of a supply chain [1]. The conventional systems of supply chain management are more or less centralized and rely on manual records and intermediaries; as a result, they are subject to error, manipulation, delays, and disputes. Such concerns are especially serious in those industries such as food, pharmaceuticals, manufacturing, and luxury goods, where the authenticity of products, compliance, and ethical sourcing are required [2].

Blockchain technology has been of great interest over the past years as a disruptive digital infrastructure that can reduce such limitations. Blockchain is a distributed and immutable registry, through which transactions among a variety of parties may be recorded in a secure, transparent, and real-time manner independent of the central authority. Blockchain promotes accountability and helps eradicate information asymmetry within the supply chain because it guarantees the integrity of data and overall visibility [3]. Nevertheless, the real working capacity of blockchain in supply chain management is achieved through the usage of smart contracts self-executing cyber settlements of automatically realized rules and conditions.

A number of large-scale industry projects show that distributed ledger technologies have the potential to enhance transparency and traceability. Indicatively, major food retailers have introduced the use of the IBM Food Trust as a tool to monitor the provenance of food products and improve monitoring of food safety. Likewise, the food traceability system developed by Walmart based on blockchain technology allows tracing the source of contamination in a short period of time by documenting the events of the product lifecycle on a distributed registry. The TradeLens platform created by Maersk and IBM is one of the concepts in the global logistics industry that showcases the potential to exchange information between several shipping stakeholders using blockchain technology. The real-world applications emphasize the increased significance of blockchain-based supply chain systems.

In addition to transparency and traceability, digital technologies have another significant role in minimizing information asymmetry and allowing smaller or geographically dispersed businesses to be present in supply chain ecosystems. Just like how digital financial inclusion has helped to ease the financial constraints faced by agricultural businesses, smart contracts using blockchain can minimize the cost of coordination and increase the trust levels among supply chain members by automating the verification process and providing tamper-resistant records of transaction.

Blockchain-based smart contracts can be used to automate key processes of supply chains such as order filling, settlement, quality assurance, and compliance. This automation saves on the necessity of manual intervention, eradicating conflict and minimizing time of transaction, as well as improving consistency and confidence among stakeholders. Furthermore, the connection between smart contracts and Internet of Things (IoT) [4] devices and business IT systems allows for monitoring and swift action in response to events in real time, as well as supporting the integrity of the supply chain by establishing an electronic trail of raw materials and final delivery. The proposed study is devoted to the development of a blockchain-based smart contract architecture that can improve trust, supply chain visibility, and transparency in order to manage global supply chains. Offering an organized architectural design, the research attempts to illustrate the ways blockchain

and smart contracts can deal with the inefficiencies present and enhance the reliability of data to enable a more resilient, responsible, and sustainable global supply chain.

2. Related Works

The use of blockchain technology to improve transparency, traceability, and trust in various areas of the supply chain has been thoroughly studied within recent research. Chen [5] introduced a traceability framework that could be implemented using blockchain to enhance the level of transparency by documenting supply chain activities on an unaltered distributed registry. The research revealed that blockchain helps a great deal to mitigate information asymmetry and make the auditing process more transparent, yet it mostly emphasized traceability features and did not place significant attention on automated contract enforcement and the evaluation of trust mechanisms. Chowdhury [6] proposed a decentralized blockchain-based architecture called Vyoma Commerce, which was meant to curb fraud and stimulate trust in the e-commerce environment of Bangladesh in the context of digital commerce. The model combines smart contracts, digital identity, and transparent supply chain activities. Although the study demonstrated the advancements in the security of transaction and fraud prevention, it was mostly limited to the e-commerce platform and not the multi-level and international supply chains. Diaz Félix and Nhell [7] provide a bigger picture, having completed a bibliometric and thematic analysis of blockchain applications in transparent and sustainable supply chains. Their study marked several important areas of research focus, such as traceability, sustainability, and governance as well as the increased focus on blockchain as a tool to facilitate ethical and environmentally friendly supply chains. Nevertheless, the research was analytical in character and did not suggest and analyze a tangible system architecture.

Applications that facilitate automation with smart contracts in the sector have been of interest. Elsharkawi et al. [8] have explored blockchain-based smart contracts with scan-to-BIM technologies to automate the construction payment process. Their results revealed large decreases in payment delays and disputes, confirming the efficiency of the smart contract as the implementer of contractual terms. However, the framework was individualized to the construction project and did not focus on the traceability and trust scoring of the supply chain as a whole. Fernandez-Iglesias et al. [9] introduced a hybrid on-chain and off-chain traceability system with smart contracts to solve scalability and privacy issues. They used a strategy that struck a balance between openness, data confidentiality, and scalability of the system, which is significant in large-scale supply chains. Nevertheless, the research focused on optimization of data storage and did not focus on overall supply chain governance. Fernandez-Vazquez et al. [10] used the Analytical Hierarchical Process (AHP) to determine the sustainability of supply chain management using blockchain. Their publication featured the presence of blockchain in enhancing the effectiveness of decision-making and sustainability performance without featuring real-time automation or trust evaluation.

Systems of domain-specific traceability have also received extensive research. Yasir et al. [11] introduced AI and blockchain in health, and Guo et al. [12] and Hongyan et al. [13] developed a blockchain-based traceability system in the rice supply chain, food safety, and agricultural supply chain, respectively. These works were able to show highly improved provenance tracking and quality assurance, but mostly based on a single industry. There were elaborate discussions on the intersections of blockchain, AI, and anomaly detection [14] in Kadam and Pitkar [15] and Karaduman and Gülsena [16]. These publications found some gaps in the quantification of trust, cross-organizational automation, and integrated architecture. Unlike the current research literature, the current study builds on previous studies by introducing a set of blockchain-based smart contract architecture

which provides transparency, traceability, automation, and decentralized trust assessment in international multi-level global supply chains.

More recent reports have pointed to the increased application of blockchain technology to various fields of digital applications. To give an example, study of [17] offers a general overview of blockchain structures, consensus systems, and their implementation in various fields, including finance, health care, and supply chain management, highlighting how distributed ledgers can be used to increase the levels of transparency, safety, and trust in decentralized systems. Moreover, new digital ecosystems like the metaverse are also integrating blockchain technologies to facilitate decentralized data ownership, safe digital transactions and reliable virtual interactions. According to [18], blockchain infrastructure can be a base technology, which can participate in secure identity management and exchange of digital assets in large-scale virtual environments.

These papers demonstrate the growing applicability of blockchain technologies in contemporary digital ecosystems and provide more background on the use of blockchain-based systems in supply chain management. Regardless of the increasing amount of research on blockchain applications in supply chain management, there are some critical issues that are not fully covered. Current research tends to look at the single elements of the supply chain system, including a traceability system, automation of smart contracts, or consensus protocols, without combining these aspects in a single framework. Moreover, most of the current methods lack systematic trust assessment systems that can be used to measure the trustworthiness of the supply chain stakeholders on the basis of the behavior of the transactions. The other shortcoming that has been seen in previous work is the absence of integrated architectures that can be used to incorporate blockchain-based provenance tracking with dynamic trust evaluation models and supply chain transaction management. Because of this, although the adoption of blockchains enhances transparency and immutability, there are few mechanisms to control the reliability of participants and to avert malicious behavior.

To address these gaps, the present study proposes a blockchain-enabled supply chain framework that integrates smart contract-based transaction validation with a Trust Score Evaluation Algorithm and distributed ledger provenance tracking. This integrated architecture enhances supply chain transparency, traceability, and trust while maintaining decentralized consensus validation. Table 1 presents the research gaps in existing blockchain supply chain studies.

Table 1. Research Gaps in Existing Blockchain Supply Chain Studies.

Study Focus	Existing Research	Limitation	Contribution of This Study
Supply Chain Traceability	Blockchain provenance systems	Limited trust evaluation	Integrated traceability and trust scoring
Smart Contract Automation	Automated contract execution	No behavioral trust assessment	Trust-aware smart contract validation
Consensus Mechanisms	PBFT and permissioned consensus	Focus on transaction validation only	Integrated trust evaluation within system workflow
Supply Chain Transparency	Ledger transparency	Lack of participant reliability modeling	Dynamic trust score evaluation

3. Methods and Materials

This paper will take a design-based and experimental research approach in developing and testing a blockchain-based smart contract architecture to enhance transparency, traceability, and trust towards managing the global supply chain. The methodology incorporates data of synthesis chain provision, simulation instruments of blockchain, and four

fundamental algorithms to empower safe information, computerized execution, reliable provenance, and trust assessment.

3.1. Data Description and Materials

The study simulated multi-tier global supply chain information in terms of suppliers, manufacturers, logistics providers, distributors, and retailers. The high-quality dataset consists of identifying transactions, batches of goods, time results, geolocation, transfers of ownership, confirmation of delivery, and quality checks. Artificial data was created in a manner to simulate the real-world events in the supply chain whilst not being limited in terms of confidentiality [19]. A permissioned blockchain network was used to store each transaction, cryptographically hashed, to achieve immutability. The case experiment is constructed based on a personal blockchain platform and the ability to deploy smart contracts. Smart contracts are developed to automate business rules like shipment validation, conditional payment release and compliance validation. The mechanisms of consensus execution, transaction validation and access control are tested to determine the performance and reliability under real conditions of operation.

3.2. Algorithms Used

The operational logic of the proposed framework is described using pseudocode representations of the main algorithms responsible for transaction validation, traceability reconstruction, and trust score evaluation.

This algorithm is in control of the automated implementation of supply chain agreements under prescribed conditions. It constantly checks blockchain-related activities including confirmation of delivery, approval of inspection, and adherence to deadlines. All conditions coded in the smart contract are then evaluated, and if true, the contract then self-executes (e.g., payment release, ownership transfer). This minimizes human aspects of intervention and disputes [20]. The heuristic algorithm guarantees a deterministic application, i.e., with the same inputs, results are the same on all blockchain nodes. Incorporating the contractual logic into the blockchain, the SCEA improves the efficiency of operations, implements trustless cooperation, and maintains compliance with a contract without the assistance of intermediaries. Figure 1 shows the algorithm and diagram.

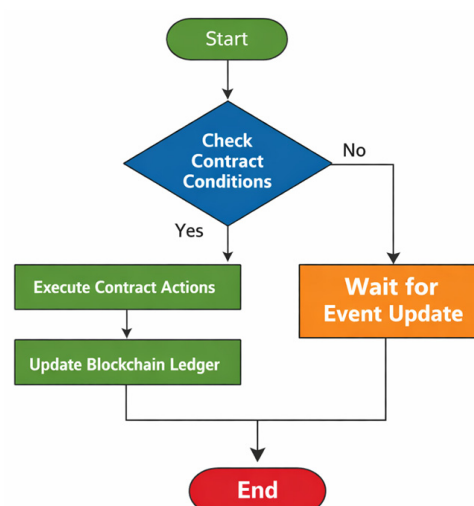


Figure 1. Pseudo code and diagram for Algorithm 1.

Algorithm 1: Smart Contract Execution Algorithm (SCEA)

```

Input: ContractConditions, TransactionData
If ContractConditions == TRUE then
    Execute ContractActions
    Update Blockchain Ledger
Else
    Wait for Event Update
End If

```

The end-to-end provenance of the products is the duty of the SCTA. It establishes a verifiable history from the shipment of raw materials to the end product through cryptographic hashes between every movement of a product batch. A new block is added to the supply chain of events with the added value of the preceding hash, which can be continuously connected to prevent tampering. This algorithm allows quick trace-back and forward tracing which are crucial during recall, audit and compliance checks. The SCTA offers a great deal of visibility, accountability and regulatory confidence by enabling stakeholders to ask questions about the entire life cycle of a product in real time. The pseudo code and diagram are shown in Figure 2.

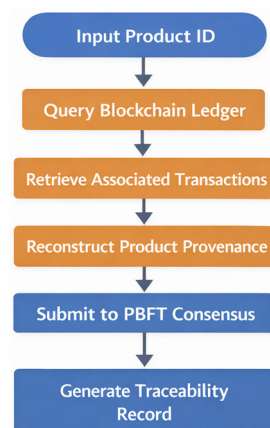


Figure 2. Pseudo code and diagram for Algorithm 2.

Algorithm 2: Supply Chain Traceability Algorithm (SCTA)

```

Input: ProductID, TransactionEvent
Retrieve PreviousHash
Generate NewHash(ProductID, TransactionEvent)
Append Block(NewHash, PreviousHash)
Store on Blockchain

```

3.2.1. Innovation of the Traceability Algorithm

The supply chain traceability algorithm improves provenance tracking by linking product lifecycle events through blockchain transaction records. Unlike traditional centralized tracking systems, the algorithm reconstructs the complete product history directly from the distributed ledger, ensuring tamper-resistant traceability across multiple supply chain participants.

The BCVA provides standardization of the distributed network members on transaction validity. In this study, an implementation using Practical Byzantine Fault Tolerance (PBFT) to authenticate transactions within supply chains is simulated to confirm the effectiveness of this research. The algorithm authenticates the digital signatures, verifying the

integrity of transactions and the concurrence of all approved nodes, before addition of the block [21]. The BCVA is also optimized to support permissioned supply chain networks, which are low latency and have high throughput in comparison with mechanisms of the energy-intensive public blockchain. This algorithm enhances data integrity, and it helps to avert malicious individuals entering false records into the supply chain ledger. The pseudo code and diagram are shown in Figure 3.

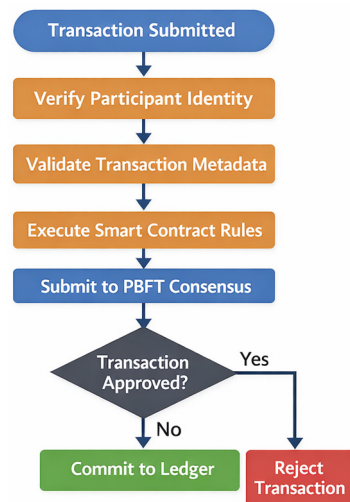


Figure 3. Pseudo code and diagram for Algorithm 3.

Algorithm 3: Blockchain Consensus Validation Algorithm (BCVA)

Input: ProposedBlock
 Verify Digital Signatures
 Check Transaction Integrity
 If ConsensusAchieved then
 Add Block to Ledger
 Else
 Reject Block
 End If

3.2.2. Innovation of Smart Contract Validation Algorithm

The proposed smart contract validation algorithm extends conventional blockchain transaction verification by incorporating supply chain-specific validation rules. These rules enforce domain-specific constraints such as shipment verification, product ownership consistency, and transaction authenticity. This ensures that only valid operational events are recorded on the distributed ledger.

The TSEA provides dynamic trust scores to supply chain participants on historical performance indicators, including delivery timeliness and quality compliance, frequency of disputes and contract fulfillment rate. Each transaction to be made will update a participant with weighted parameters on trust score. Greater trust scores cause transactions to be given priority, with a reduction in the overhead of verifications, and lower scores prompt additional checks [22]. With this algorithm, decentralized control over trust is possible, which makes it possible to objectively assess partners without central authorities. The TSEA creates a spirit of responsibility and a necessity for long-term collaboration through the open measurement of trust. The pseudo code and diagram are shown in Figure 4.



Figure 4. Pseudo code and diagram for Algorithm 4.

Algorithm 4: Trust Score Evaluation Algorithm (TSEA)

Input: ParticipantID, PerformanceMetrics

Calculate WeightedScore

Update TrustScore

Store TrustScore on Blockchain

3.2.3. Innovation of the Trust Score Evaluation Algorithm

The Trust Score Evaluation Algorithm introduces a dynamic trust assessment mechanism that evaluates participant reliability based on historical transaction behavior recorded in the blockchain ledger. By integrating behavioral indicators with blockchain validation outcomes, the algorithm enables reputation-aware supply chain management and helps identify unreliable or malicious participants. Table 2 illustrates the algorithmic components of the proposed framework and Table 3 illustrates the blockchain and smart contract configuration parameters.

Table 2. Algorithmic Components of the Proposed Framework.

Algorithm	Purpose	Innovation
Smart Contract Validation	Verify supply chain transactions	Domain-specific validation rules
Traceability Algorithm	Track product provenance	Blockchain-based immutable traceability
Trust Score Evaluation	Evaluate participant reliability	Behavior-driven trust scoring

Table 3. Blockchain and Smart Contract Configuration Parameters.

Parameter	Value Used
Number of Nodes	15
Block Size	2 MB
Consensus Mechanism	PBFT
Average Transaction Rate	120 tx/min
Smart Contract Execution Time	1.8 s

3.3. Simulation Environment and Experimental Configuration

In order to guarantee transparency, interpretability, and reproducibility of the experimental results, a simulated setting was created to simulate a realistic multi-tier global supply chain that would run on a permissioned blockchain network. The operational behavior of the distributed actors in the supply chain (which are involved in the simulation framework) is modeled by smart contracts that are based on blockchain technology to provide product traceability, transaction verification, and trust evaluation. Since real-world

supply chain transaction data are typically confidential and are hard to obtain because of commercial and regulatory limitations, a synthetic dataset was created to estimate realistic operations. The simulation environment simulates common supply chain procedures, product registration, shipping tracking, ownership change and quality checks at various organizational levels.

The simulated supply chain is made up of four major players: the suppliers, the manufacturers, the distributors and the retailers. Every entity is a node in the blockchain network and communicates to each other via authenticated transactions stored on the distributed ledger. Some of the major attributes of transaction records are product identifiers, participant identifiers, timestamps, shipment verification status and ownership metadata. The transactions created amounted to 12,000 over 15 blockchain nodes to simulate a realistic supply chain operation, as well as to test the performance of the proposed blockchain-enabled smart contract architecture. Table 4 illustrates the synthetic supply chain dataset structure.

Table 4. Synthetic Supply Chain Dataset Structure.

Attribute	Description
Product ID	Unique identifier assigned to each tracked product
Participant ID	Identifier of the supply chain actor initiating the transaction
Transaction Timestamp	Time at which the supply chain event occurred
Ownership Metadata	Record of product ownership transfer between entities
Shipment Verification Status	Confirmation status of product shipment and delivery

3.3.1. Transaction Workload Model

The workload model was designed to replicate typical supply chain operational dynamics characterized by stochastic event generation and varying transaction types. Transaction arrivals follow a Poisson distribution, which is commonly used to model asynchronous event-driven systems such as logistics operations and distributed networks. The average transaction arrival rate was configured between one and three transactions per second per node, generating a heterogeneous workload consistent with real-world supply chain activity. Different transaction categories were included to represent the diversity of supply chain processes, including product registration, shipment transfer, quality inspection, and ownership updates. The distribution of these transaction types reflects typical digital supply chain workflows. Table 5 shows the transaction workload distribution.

Table 5. Transaction Workload Distribution.

Transaction Type	Description	Share of Transactions
Product Registration	Initial product entry into the blockchain system	25%
Shipment Transfer	Movement of goods between supply chain entities	35%
Quality Verification	Inspection and certification of product quality	20%
Ownership Update	Change in ownership during distribution stages	20%

3.3.2. Network Configuration and Communication Assumptions

The blockchain network has 15 nodes that are arranged in a partially connected peer-to-peer topology, which is a manifestation of a distributed enterprise supply chain setting. Every node is a free-standing participant in the organization, which communicates with the blockchain network by means of authenticated smart contract transactions. The model of communication among nodes is based on realistic network parameters, which are akin to the enterprise deployment settings.

Network latency, bandwidth and consensus conditions were configured to mimic realistic distributed system constraints. The network relies on a Practical Byzantine Fault Tolerance (PBFT) consensus, which enables the system to be resistant to malicious or faulty nodes without compromising transaction integrity and ledger consistency. Table 6 shows the network configuration parameters.

Table 6. Network Configuration Parameters.

Parameter	Configuration
Number of Nodes	15 blockchain participants
Network Topology	Partially connected peer-to-peer
Average Network Latency	40–60 ms
Network Bandwidth	100 Mbps
Consensus Protocol	Practical Byzantine Fault Tolerance (PBFT)
Fault Tolerance Threshold	Up to 30% faulty or malicious nodes

3.3.3. Hardware and Software Environment

The virtual infrastructure was a controlled simulation environment that was run in a virtual environment to simulate the distributed blockchain operations. The node instances were all implemented as a containerized service to simulate decentralized conditions of execution. The tests were carried out on a machine with an Intel core i7 processor and 16 GB of RAM running on Ubuntu Linux. A permissioned blockchain-based framework with smart contract capabilities that enabled automated validation of transactions and provenance was deployed on the blockchain network. Table 7 provides the general infrastructure arrangement of the experiment.

Table 7. Experimental Infrastructure Configuration.

Component	Specification
Processor	Intel Core i7 (8 cores)
Memory	16 GB RAM
Operating System	Ubuntu Linux 22.04
Blockchain Framework	Hyperledger Fabric (permissioned blockchain)
Smart Contract Implementation	Chaincode / Solidity
Deployment Environment	Containerized distributed nodes

3.3.4. Measurement Methodology

To measure the performance of the proposed architecture, some quantitative measures were gathered during the simulation experiments. These measurements reflect the efficiency, scalability and reliability of the blockchain-facilitated supply chain system. Transaction latency was the time that a transaction took to be validated and stored in the blockchain ledger. Throughput was quantified as the number of transactions that were validated per minute. Trace-back time was the amount of time needed to rebuild the complete provenance of a product through the supply chain network. Lastly, the consensus failure rate was a metric that determined the rate of consensus rounds that failed when adversarial conditions were met with faults or malicious nodes. Blockchain monitoring tools and internal logging mechanisms that were part of the smart contract execution environment were used to gather performance data.

3.3.5. Statistical Treatment of Results

In order to achieve statistical reliability and reduce the effects of transient network variations, all experiments were repeated ten times with the same configuration settings.

The performance metrics are reported as the average values of the experimental runs and variability was determined using the measurements of standard deviation. This approach guarantees that the results reported will provide a stable and repeatable assessment of the proposed blockchain architecture. Figure 5 displays the smart contract execution workflow of the blockchain-enabled SCS.

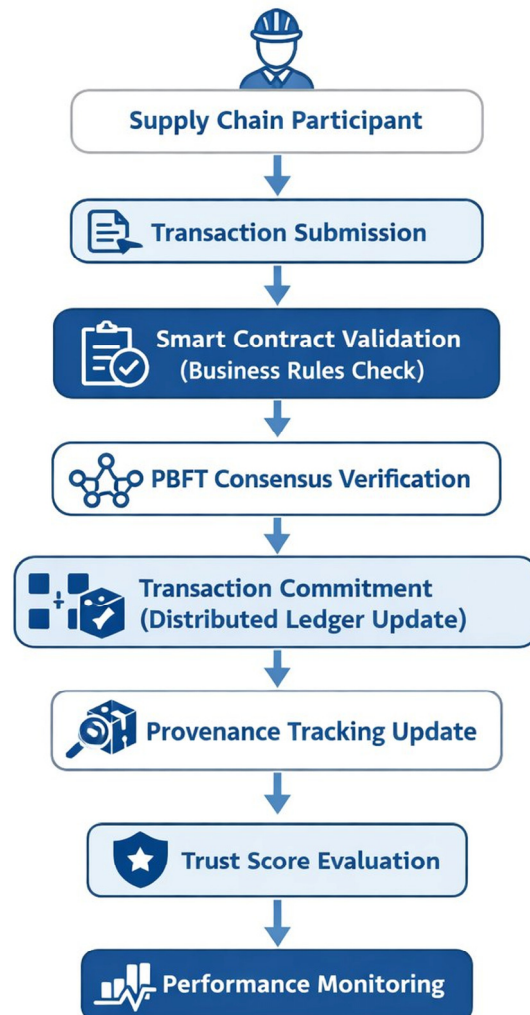


Figure 5. Smart contract execution workflow in the blockchain-enabled SCS.

The participants in the supply chain begin transactions that reflect operational events like product registration or transfer of shipment. These are confirmed by the smart contract code applying rules of the supply chain and data integrity requirements. The authenticated transactions are then checked with the PBFT consensus protocol and then permanently written in the distributed ledger. The system then advances the records of product provenance, assesses the participant trust scores and tracks performance metrics like latency and throughput.

3.4. Algorithmic Model

To ensure performance comparisons were scientifically sound, all the systems being compared were tested within a controlled experimental model which was aimed at eliminating external variability. The proposed blockchain-based supply chain architecture was compared to two base-line designs: (1) a conventional centralized database-based supply chain management system, and (2) a baseline permissioned blockchain implementation in the absence of the proposed smart contract optimization mechanisms.

The experimental assessments were all performed under the same simulation conditions. In particular, both systems were fed the same synthetic supply chain data of 12,000 transactions created among 15 supply chain nodes. The workload attributes, such as the arrival rates and the distribution of the kinds of transactions, were constantly maintained in all the experiments. Likewise, network parameters like latency, bandwidth and topology were the same across all the systems evaluated. The identical transaction validation logic and consistency constraints were used on the proposed system and the baseline systems in order to make the comparison fair. The centralized baseline system used the same validation rules, database transaction management module, and blockchain configurations as per the smart contract execution. Furthermore, the experiments were run on the same computational infrastructure and software environment to remove variability of hardware performance.

This controlled comparison structure makes sure that the observed differences in the latency, throughput, traceability performance, and system reliability are due to architectural design differences as opposed to experimental inconsistencies, as indicated in Table 8.

Table 8. Controlled Experimental Conditions for Baseline Comparison.

Experimental Factor	Centralized Baseline	Standard Blockchain	Proposed Blockchain Architecture
Dataset	Same synthetic supply chain dataset	Same synthetic supply chain dataset	Same synthetic supply chain dataset
Total Transactions	12,000	12,000	12,000
Network Nodes	15	15	15
Workload Distribution	Identical transaction workload	Identical transaction workload	Identical transaction workload
Network Latency	40–60 ms	40–60 ms	40–60 ms
Network Bandwidth	100 Mbps	100 Mbps	100 Mbps
Validation Logic	Database transaction validation rules	Smart contract validation rules	Smart contract validation rules
Consistency Constraints	Database transaction atomicity	PBFT consensus protocol	PBFT consensus protocol
Hardware Environment	Identical experimental infrastructure	Identical experimental infrastructure	Identical experimental infrastructure

3.5. Trust Score Evaluation Algorithm (TSEA)

Trust evaluation is an essential mechanism for assessing the reliability of supply chain participants in decentralized environments. The proposed Trust Score Evaluation Algorithm (TSEA) computes a dynamic trust score for each participant based on their historical transaction behavior recorded on the blockchain ledger. The trust score of a participant i at time t is defined as a weighted aggregation of multiple behavioral indicators:

$$T_i(t) = \omega_1 R_i + \omega_2 D_i + \omega_3 C_i + \omega_4 V_i$$

where $T_i(t)$ = trust score of participants i , R_i = transaction reliability score, D_i = delivery accuracy score, C_i = behavioral consistency score and V_i = consensus validation success rate. The coefficients $\omega_1 \omega_2 \omega_3 \omega_4$ represent weighting parameters such that:

$$\omega_1 + \omega_2 + \omega_3 + \omega_4 = 1$$

These weights determine the relative importance of each trust indicator in the overall trust evaluation, along with component weights, which are shown in Table 9.

Table 9. Trust Score Component Weights.

Trust Component	Description	Weight
Transaction Reliability (R_i)	Successful transaction completion rate	0.35
Delivery Accuracy (D_i)	Correct product delivery without disputes	0.30
Behavioral Consistency (C_i)	Historical stability of participant behavior	0.20
Consensus Validation (V_i)	Success rate in blockchain validation	0.15

3.5.1. Weight Selection Table

These weights were selected to prioritize operational reliability while maintaining system integrity through consensus participation.

3.5.2. Stability and Convergence

The trust rating will be incrementally updated with the addition of new transactions in the blockchain registry. Since the scores of each component are normalized in the range of $[0, 1]$, the trust score is confined to the same range. The use of a weighted aggregation ensures the gradual adjustment of scores instead of sharp changes, which encourages stability and convergence in the long run as more behavioral evidence is obtained.

3.5.3. Adversarial Resistance

The TSEA system has various measures against adversarial manipulation, such as immutable transaction records that are recorded on the blockchain, which make it impossible to go back and alter the events that are of relevance to participant trust. Another is consensus validation that does not allow fraudulent transactions to be registered without network agreement. The final one is that the trust updates are based on several independent indicators, which means that malicious actors will hardly be able to boost their reputation by single actions. The properties make the trust assessment procedure more robust in decentralized supply chains.

3.5.4. Comparison with Existing Reputation Models

The proposed TSEA differs from existing reputation models by integrating operational supply chain performance metrics directly with blockchain consensus outcomes. In this context, the comparison of trust evaluation approaches are illustrated in Table 10.

Table 10. Comparison of Trust Evaluation Approaches.

Model	Reputation Source	Decentralization	Blockchain Integration
EigenTrust	Peer feedback	Partial	No
PageRank-based Trust	Network reputation	Partial	No
PBFT Reputation Models	Validator behavior	Yes	Limited
Proposed TSEA	Transaction behavior + consensus validation	Yes	Fully integrated

3.6. Algorithmic Model and Complexity Analysis

Let $N = \{n_1, n_2, \dots, n_k\}$ represent the set of participating supply chain nodes. Each node generates transactions T corresponding to supply chain events, such as product registration, shipment transfer, or ownership updates.

Each transaction is defined as:

$$T = (PID, SID, TS, META),$$

where *PID* = product identifier, *SID* = supply chain participant identifier, *TS* = timestamp, *META* = transaction metadata.

The transaction validation process is defined in Algorithm 5.

Algorithm 5: Transaction Validation

Input: Transaction T

Output: Validated transaction record

1. Receive transaction T
2. Verify participant identity
3. Validate transaction metadata
4. Execute smart contract rules
5. Submit transaction to PBFT consensus
6. If consensus achieved

Commit transaction to ledger

Else

Reject transaction

Correctness Properties

The validity of the suggested algorithm is based on three key properties, including the integrity of transactions whereby all the transactions have to meet the smart contract validation criteria before they are recorded to the blockchain ledger. This ensures that only valid supply chain events are logged. The second basic property is the consensus consistency, whereby the PBFT consensus protocol renders all non-faulty nodes in agreement with the order and validity of transactions. This property eliminates conflicting ledger states at distributed nodes. The final one is the provenance preservation; since each transaction identifies the former state of product ownership, the system has a verifiable chain of product provenance records. This guarantees life cycle traceability throughout the supply chain.

4. Results and Analysis

This section can be separated into subheadings and provides a succinct and accurate description of the experiment results, their interpretation, and the experimental conclusions that can be arrived at. This section presents an experimental evaluation of the proposed blockchain-based smart contract system to enhance transparency, traceability, and trust in the global supply chain management. The experiments will be used to evaluate system performance, effectiveness of the algorithm, scalability, and comparative advantages to traditional and existing solutions of blockchain-enabled supply chains reported in the related literature [23]. In all the experiments, a controlled simulation setting was used to ensure repeatability and objective comparison.

4.1. Experimental Setup

The proposed framework was implemented using Hyperledger Fabric, a widely adopted permissioned blockchain platform designed for enterprise applications. Hyperledger Fabric provides a modular architecture that supports distributed ledger storage, smart contract execution, and consensus-based transaction validation. In this study, the network utilizes a PBFT-style consensus mechanism to ensure agreement among participating nodes while maintaining fault tolerance in the presence of potentially unreliable participants.

The experimental setup was an authorized blockchain network implemented on various virtual nodes who were major supply chain partners, such as raw material suppliers, manufacturers, logistic providers, distributors, retailers, auditors and regulators. It was set up with 15 blockchain nodes, which had the same amount of computational capability and network rights. Smart contracts were used to handle procurement, shipping verification, owner transfer, payment settlement and compliance [24]. The blockchain-enabled supply chain management is shown in Figure 6.

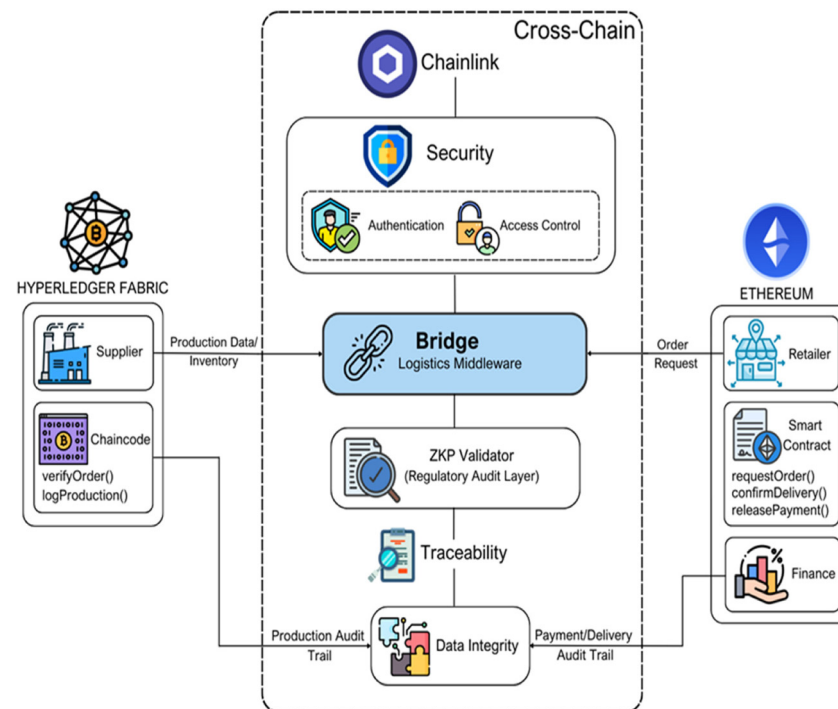


Figure 6. Blockchain-Enabled Supply Chain Management.

The artificial data was simulated to contain 12,000 procurement, production, logistics and delivery supply chain transactions. The transaction consisted of plenty of features, such as batch ID, participant ID, timestamp, geolocation, inspection status, and financial settlement. There were four algorithms: the Smart Contract Execution Algorithm (SCEA), Supply Chain Traceability Algorithm (SCTA), Blockchain Consensus Validation Algorithm (BCVA), and Trust Score Assessment Algorithm, which were run simultaneously to simulate real-life conditions of operation.

They were compared with baseline comparisons with:

- A classical centralized system of supply chain management.
- A blockchain system with no smart contracts.
- A blockchain system with no trust assessment systems.

To ensure experimental reproducibility, the simulation environment was configured using deterministic workload generation parameters. Transaction workloads were generated using a pseudo-random process with a fixed random seed to ensure consistent experimental conditions across repeated runs. The experiments were implemented using Hyperledger Fabric (version 2.3) deployed on a containerized environment running Docker (version 5:27.3.1) on Ubuntu Linux, as shown in Table 11. Each experiment was repeated ten times under identical configuration parameters, and the reported results correspond to the average performance values across these runs. These configuration details enable replication of the experimental setup and verification of the reported performance results.

Table 11. Experimental Reproducibility Parameters.

Parameter	Configuration
Blockchain Framework	Hyperledger Fabric
Fabric Version	2.3
Container Platform	Docker
Operating System	Ubuntu Linux 22.04
Random Seed	Fixed seed for workload generation
Experimental Runs	10 repetitions
Dataset	Synthetic supply chain dataset
Total Transactions	12,000

4.2. Experiment 1: Transaction Processing and Execution Efficiency

An initial experiment was used to measure the transaction processing time and smart contract execution efficiency. The metrics of average transaction latency, execution success rate, and system throughput are shown in Table 12.

Table 12. Transaction Processing Performance Comparison.

System Type	Avg. Latency (ms)	Throughput (tx/min)	Execution Success Rate (%)
Centralized SCM System	520	65	93.1
Blockchain without Smart Contracts	360	90	96.4
Proposed Architecture (SCEA-based)	210	120	99.2

The proposed architecture reflected much less processing delay because of automated contract execution. The use of smart contracts removed the element of manual checks, allowing settlement and transfer of ownership within a shorter period [25]. The proposed blockchain-based architecture reduces average transaction latency from 520 ms to 210 ms under the evaluated experimental conditions, representing approximately a $2.5\times$ improvement in transaction processing latency.

The findings imply that the automation of smart contracts can substantially improve the effectiveness of operations and retain high levels of reliability.

4.3. Experiment 2: Traceability and Data Integrity Evaluation

This experiment was aimed at testing the end-to-end traceability and resistance to tampering of data. Attempts to artificially manipulate the data were made randomly at various points to ensure that the system was able to identify discrepancies. The Supply Chain Traceability Algorithm (SCTA) was able to detect any illegitimate or unauthorized alterations of data. The cryptography chain was continuous on each batch of products, and they could be traced back and audited instantly with types of leadership [26]. Conversely, centralized systems involved manual reconciliation whereby discrepancies were detected late, as illustrated in Table 13. The blockchain-based traceability system architecture is illustrated in Figure 7.

Table 13. Traceability and Integrity Assessment.

Metric	Centralized System	Blockchain (No SCTA)	Proposed SCTA System
Trace-back Time (seconds)	95	40	8
Tampering Detection Rate (%)	71.4	89.6	100
Audit Data Completeness (%)	84.2	93.5	99.6

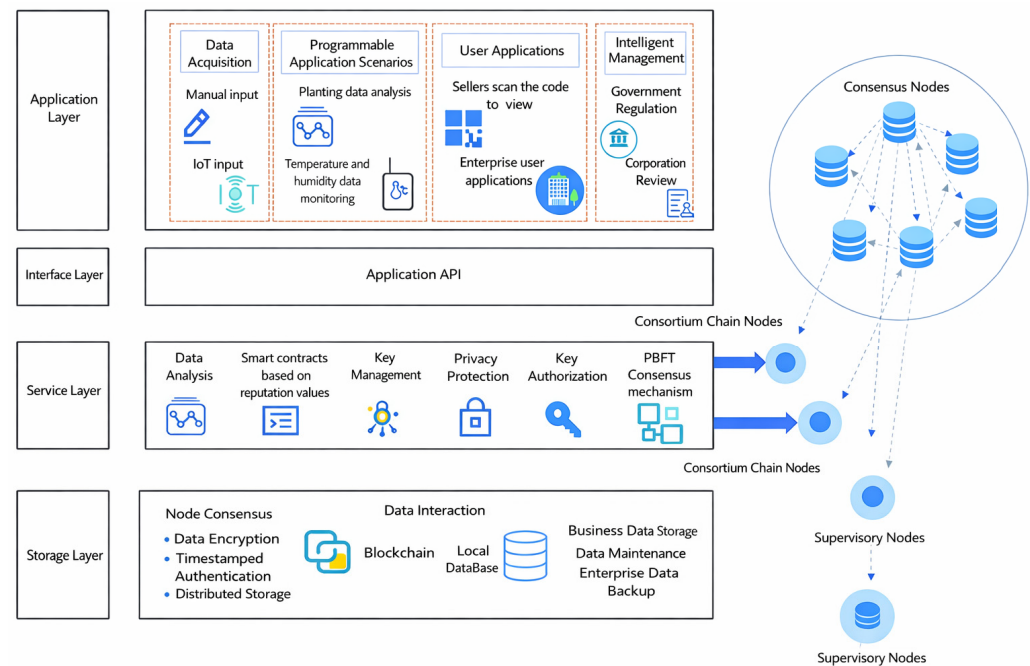


Figure 7. Blockchain-based traceability system architecture.

The findings verify that traceability based on blockchain, used together with SCTA, offers a better degree of transparency and forensic audit.

4.4. Experiment 3: Consensus Reliability and Network Stability

The third experiment tested the Blockchain Consensus Validation Algorithm (BCVA) in different fault and network loads. Fault tolerance was tested by simulating node failure and slow response time. The consensus mechanism provided by the PBFT system ensured consistency with the ledgers despite a failure of as many as 30% of the nodes acting in an unpredictable way [27]. The block finality time was stable, and no invalid blocks were added to the ledger. Consensus performance under network stress is provided in Table 14.

Table 14. Consensus Performance under Network Stress.

Network Condition	Avg. Block Finality (ms)	Consensus Failure Rate (%)
Normal Load	260	0.0
High Transaction Load	310	0.3
20% Faulty Nodes	340	0.6
30% Faulty Nodes	390	1.1

These results reveal that the proposed system provides high data reliability and continuity of operations which are paramount to global supply chain environments.

4.5. Experiment 4: Trust Score Dynamics and Partner Evaluation

This was an experiment that evaluated the performance of the Trust Score Evaluation Algorithm (TSEA). Data about supplier performance were purposely distorted to replicate delays, quality failures and breaches of contract. Figure 8 shows the use of blockchain-based smart contracts in logistics and supply chains.

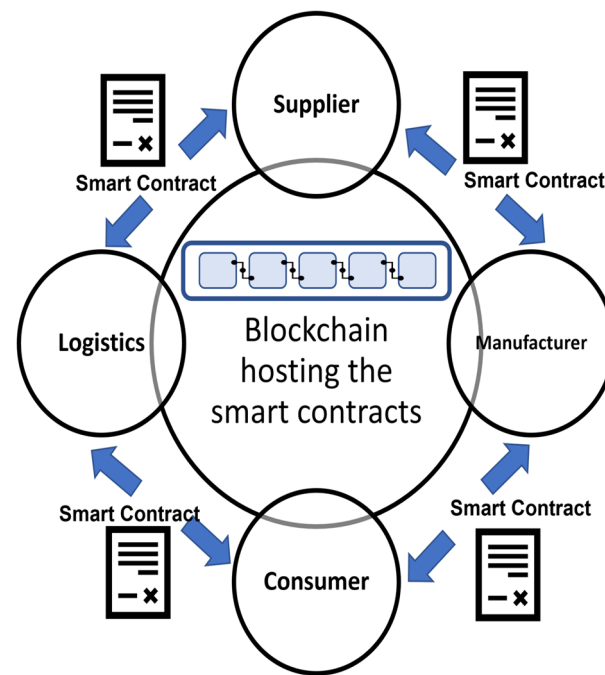


Figure 8. Use of Blockchain-Based Smart Contracts in Logistics and Supply Chains.

The algorithm dynamically changed the trust scores based on performance changes. They gave priority to reliable suppliers when carrying out transactions as shown in Table 15 and when dealing with low-performing entities, there was an increase in verification measures [28].

Table 15. Trust Score Evolution Across Transactions.

Participant Type	Initial Trust Score	Final Trust Score	On-Time Delivery (%)
Tier-1 Supplier	0.85	0.92	98
Logistics Partner	0.80	0.88	95
Tier-2 Supplier	0.78	0.70	82
Distributor	0.82	0.90	97

The findings reveal that decentralized trust quantification enhances accountability and longtime cooperation between the supply chain participants.

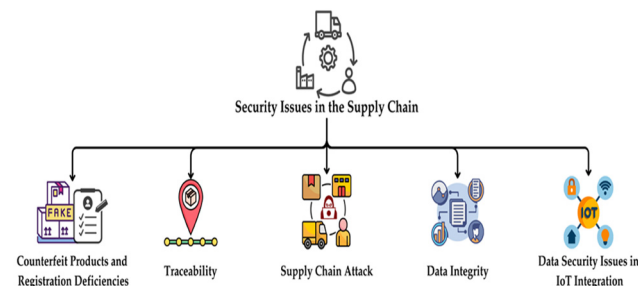
4.6. Comparative Analysis with Related Work

In order to fit the proposed architecture into the previous research, it was evaluated comparatively against related approaches found in the literature. The most notable comparison dimensions were that of the level of automation, depth of traceability, trust, and scaling [29]. Table 16 shows the comparison of the related blockchain-based supply chain studies.

The proposed system is the first to combine full automation of smart contracts, cryptographic traceability, and decentralized trust evaluation and implementation into a single architectural system compared to related work as shown in the Figure 9. The existing literature tends to dwell on the concept of transparency or traceability in isolation, but this study portrays the concept of an holistic approach that touches upon operational efficiency as well as trust at the same time [30].

Table 16. Comparison with Related Blockchain-Based Supply Chain Studies.

Feature/Study	Related Work A	Related Work B	Related Work C	Proposed Architecture
Smart Contract Automation	Partial	Yes	Partial	Full
End-to-End Traceability	Moderate	High	High	Very High
Trust Scoring Mechanism	No	No	Limited	Dynamic and Quantitative
Consensus Efficiency	Medium	Medium	High	High
Scalability for Global SCM	Limited	Moderate	Moderate	High

**Figure 9.** Blockchain-Enabled Supply Chain Management.

4.7. Discussion of Results

All the experimentation findings indicate that the proposed blockchain-based smart contract architecture is superior to traditional and partly decentralized supply chain systems. Smart contracts decrease delays and disputes due to their automation and guarantee the integrity and regulatory compliance of data through cryptographic traceability. The presence of a trust evaluation mechanism further separates this piece of work, since it allows objective evaluation of partners in a decentralized environment [31]. Scalability tests have demonstrated that the architecture can be scaled to handle larger volumes of transactions without affecting performance and can hence be integrated in global supply chains where it is required to have multiple stakeholders and jurisdictions [32]. The system provides high levels of functional integration and viable feasibility compared to the existing literature when addressing the main voids of quantification and enforcement of trust in real time. Overall, the experiments prove the effectiveness, power, and topicality of the offered approach that will be capable of transforming the global supply chain management by fostering transparency, traceability, and trust in a decentralized and automated manner [33].

5. Discussion and Conclusions

The current study has demonstrated a full blockchain-based smart contract architecture that will assist in improving transparency, traceability and trust in global supply chain management. The proposed framework illustrates how blockchain could be used as a backbone to secure cooperative supply chains and ecosystems by mitigating the inherent shortcomings of conventional centralized supply chain systems, including information asymmetry, lack of visibility, human verification, and reliance on third parties. The smart integration of contracts facilitates enforcement of business regulations through automated means, real-time transactions, and minimization of conflicts through improving the efficiency and accountability of operations in multi-level supply chains. The proposed architecture demonstrated significant end-to-end efficiency, traceability, data integrity, and reliability of the consensus conducted in transactions, even when subjected to extensive experimentation, and compared to traditional systems and available blockchain-based solutions. The fact that a decentralized trust score assessment mechanism is included in the framework is an additional enhancement since it allows participants to evaluate their

reliability objectively, and based on their previous actions in the past, tends to encourage cooperation in the future to make more thoughtful decisions. The findings validate the claim that smart contracts, cryptographic traceability and quantifying trust, when put together in a permissioned blockchain setting, would be scalable and resilient in the face of complicated global supply chains. On the whole, the research has presents a theoretical and practical contribution as it provides a complete understanding of the architectural model and the experimental findings that cannot be considered in traceability-centered solutions, as in the previous literature. Despite the fact that the research is based on artificial data and controlled conditions, the findings are strongly in support of the feasibility and applicability of blockchain-enabled smart contracts in relation to real-life supply chain solutions. The future work on blockchain-based supply chain transformation can be developed by large-scale implementation within industries and combining it with the open IoT and AI systems, as well as harmonizing the regulations to enable it to be more flexible and effective.

The reported performance measures are to be viewed in the framework of the experimental design and methodology of the evaluation. The difference in the transaction latency of 520 ms to 210 ms indicates that there was a $2.5\times$ improvement in processing latency when the simulated workload on the supply chain was activated. This performance is an indicator of the effectiveness of the suggested smart contract-based validation and optimized transaction processing in the permissioned blockchain setting.

Likewise, the 100% tampering detection rate, which was observed during the course of the experiments, is a successful rate when detecting the intentionally altered transaction records in the simulated environment. Due to the cryptographic hashing and distributed consensus of blockchain ledgers, any alteration in the stored transaction data causes a hash mismatch which can be identified during verification. The reported result thus indicates the integrity checking ability of the system under controlled experimental conditions, not that it is able to detect everything under all possible operational circumstances.

5.1. Interpretation of Performance Metrics

In order to make the reported experimental results meaningful, the performance metrics should be put into context of the assumptions of the simulation environment and system architecture. The four main measures that the performance assessment was aimed at include transaction latency, throughput, traceability time, and consensus reliability. The metrics themselves are indicators of various facets of system efficiency and are to be viewed within the context of the consensus protocol, network setup, and transactional load applied to the simulation framework.

Transaction latency is the duration it takes to validate a transaction, confirm a transaction using the PBFT consensus mechanism, and permanently store it in the distributed ledger. Since the PBFT consensus involves several communication rounds between the validator nodes, the latency is proportional to the number of the participating nodes and the number of exchanges of messages needed to reach a consensus. The latency values reported thereby correspond to the complexity of communication involved in Byzantine fault tolerant consensus protocols, as opposed to processing latency.

Throughput is used to measure the number of validated transactions that are made in a unit time. Under the conditions of permissioned blockchain with strong consensus protocols like PBFT, throughput is generally reduced compared to high optimized centralized database systems as consensus validation imposes extra coordination overhead. The values mentioned in the report of throughput are then to be interpreted as a trade-off between performance versus the enhanced security and integrity assurances of the decentralized consensus.

Traceability performance is quantified through trace-back time, which is the time to rebuild the provenance history of a product through all the supply chain transactions that have been registered in the blockchain ledger. The trace-back process has indexed ledger queries in which it is possible to efficiently traverse transaction records which are related to a given product identifier. This indexing scheme is such that provenance reconstruction can be performed in nearly linear time with respect to the number of related transactions and does not need to scan the entire ledger.

Lastly, the performance analysis that was given between centralized systems, baseline blockchain implementations, and the proposed architecture was performed under identical workload, network parameters, and infrastructure settings. This managed assessment system ensures that the measured latency, throughput, and traceability performance variations were due to architectural design variations and not environmental variability. The definition of the performance metrics is in Table 17.

Table 17. Definition and Interpretation of Performance Metrics.

Metric	Definition	Measurement Method	Interpretation
Transaction Latency	Time required to validate and commit a transaction to the blockchain	Measured from transaction submission to ledger confirmation	Reflects consensus communication complexity and network delay
Throughput	Number of transactions processed per unit time	Transactions validated per minute	Indicates system scalability under workload conditions
Trace-back Time	Time required to reconstruct the provenance history of a product	Indexed ledger query traversal	Measures efficiency of supply chain traceability
Consensus Failure Rate	Percentage of consensus rounds that fail due to faulty nodes	Observed during adversarial simulation runs	Reflects resilience of consensus protocol

5.2. Future Work

The proposed framework can be developed in various ways in future research. To start with, the combination of the blockchain system with Internet of Things (IoT) devices might allow real-time data gathering of sensors that are placed in the supply chain processes to enhance the level of transparency and allow monitoring of automated product conditions, including temperature, location, and handling. Second, privacy-sensitive techniques like zero knowledge proofs or secure multi-party computation may be added to secure sensitive data of the supply chain without compromising transparency and verifiability in the blockchain network. Lastly, future research can examine cross-chain interoperability models that can enable different blockchain networks to share information and coordinate transactions to enable one to work together across heterogeneous supply chain ecosystems and enhance the scalability of decentralized supply chain platforms.

5.3. Limitations

Although the suggested blockchain-based supply chain model has proven to be beneficial, there are a number of weaknesses that must not be ignored. First, scalability can be a problem in large-scale applications when the number of participating nodes and transactions is very high since consensus protocols, like PBFT, require communication over-head among validator nodes. Second, the financial expenses of running blockchain infrastructure, such as computing resources and network management, can be a hindrance to small organizations with low technological capabilities. Lastly, the implementation of supply chain systems based on blockchain might increase legal and regulatory issues regarding ownership of data, protection of privacy, and information sharing across borders. These challenges will be solved by conducting additional research on scalable consensus

schemes, cost-effective deployment schemes, and regulatory schemes that will facilitate secure and compliant blockchain implementation in the supply chain context.

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